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# A transparent financial risk factor library

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**Abstract**

Financial risk factor time series are essential for empirical asset pricing research, yet existing factor libraries offer limited reproducibility, lack modularity, and target single programming languages. We present a fully replicable, multilingual financial risk factor library that addresses these shortcomings. Our modular framework covers the full pipeline from raw data to factor returns: data acquisition, cleaning, variable construction, and portfolio aggregation. The library provides both an application programming interface (API) and native implementations in Python and R. A transparent, modular design supports reproducibility and flexibility across diverse portfolio sort configurations and adjustment methodologies. All code and data workflows are publicly available to promote standardization and reproducible research practices in empirical asset pricing.

*Keywords*: financial risk factors, reproducible research, asset pricing, API, R, Python.

## 1. Introduction: Constructing financial risk factors

In empirical finance, a key question is which sources of systematic risk drive asset returns. Researchers address this question by constructing factor returns: portfolios that isolate single economic characteristics or exposures. Factor returns serve model evaluation, hypothesis testing, benchmarking, and decomposition of returns into systematic and idiosyncratic components (Fama and French 1993; Cochrane 2011; Harvey, Liu, and Zhu 2016).

Despite their importance, no widely adopted, transparent, and accessible source standardizes factor construction. Current practice relies on ad-hoc publication of factor series, usually without code or detailed workflows, limiting reproducibility, comparability, and extension. Reproducible research requires transparent workflows, standardized data processing, analysis-

ready series, and modularity to vary methodological choices.[1]

Our project makes four contributions. First, we release a public factor library that centralizes series and workflows. Second, we provide a modular, end-to-end workflow for data acquisition, cleaning, variable construction, and portfolio aggregation. Third, we offer native R and Python implementations and a programmatic API for standardized data retrieval. Fourth, we publish all code, workflows, and documentation to enable replication, methodological comparison, and extensibility.

We provide two libraries, *tidyfinance* (on CRAN) and *py-tidyfinance*, that streamline data acquisition from WRDS, Compustat, and other commonly used data providers. Automated routines handle standard lagging procedures and eliminate ad-hoc fixes. The packages embed special adjustments typical for financial data, such as delistings and price or volume corrections.

A core focus of our project is portfolio sort construction, which involves many subjective choices: breakpoints, weighting, rebalancing, handling ties, and sample inclusion. Our code treats sorting as a configurable module. We document and test all choices and implement common variants (equal- or value-weighted, double sorts, quantile/percentile breaks, overlapping portfolios).

We illustrate the full workflow in R, from raw data to factor returns, including diagnostics. Python examples in the appendix ensure parity in configuration, defaults, and outputs.

We publish a curated factor library with preprocessed time series across sorting variables, weighting schemes, and adjustments. We demonstrate how methodological choices affect factor returns to highlight their empirical relevance. A programmatic API provides access to standardized series and preprocessed panels, with support for versioning and batch retrievals. We publish all software, workflows, and documentation to enable replication, sensitivity analysis, and extension.

## 2. Vignette: A case study in factor construction

Our focus is on presenting `tidyfinance`. You can install the released version of `tidyfinance` from CRAN via:

```
install.packages("tidyfinance")
```

You can install the development version of `tidyfinance` from GitHub via:

```
# install.packages("pak")
pak::pak("tidy-finance/r-tidyfinance")
```

```
library(dplyr)
library(tidyr)
library(lubridate)
library(ggplot2)
```

[1] Existing resources illustrate the gap: the French Data Library (Fama and French 2023) provides canonical factors but no code; the Open Source Asset Pricing Project (Chen and Zimmermann 2022) focuses on replication; the global factor database (Jensen, Kelly, and Pedersen 2023) targets Python only. A modular, multilingual framework is still needed.

```
library(tidyfinance)
```

We walk through the construction of a single factor, asset growth, from raw data to long-short portfolio returns. Each step generalizes to the full library of sorting variables. Figure 1 shows the end-to-end pipeline and highlights the `tidyfinance` functions involved at each stage.

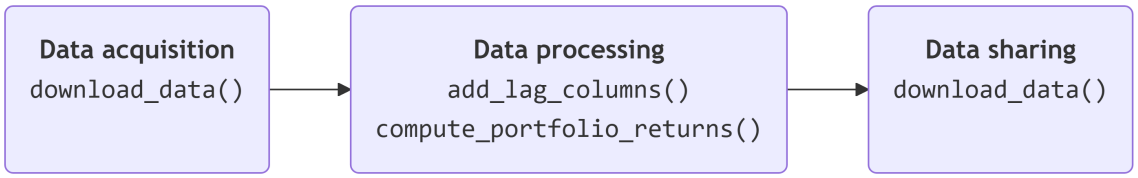



Figure 1: Research workflow for factor construction with `tidyfinance`. Each box lists the main function calls for that stage.

## 2.1. Data sources

A central design goal of `tidyfinance` is to provide a single, consistent interface, `download_data()`, to a wide range of financial and economic data sources. The vignette only requires two of them (CRSP monthly stock returns and Compustat annual financial statements), but the same function gives access to 318 dataset types spanning equity returns, fixed-income transactions, macroeconomic predictors, and more. The package serves researchers well beyond asset pricing: macro forecasting (FRED, Goyal-Welch), credit markets (FISD, TRACE), and portfolio benchmarking (index constituents) all use the same unified workflow.

For our asset growth example, we download and preprocess the two required datasets as follows. Before the first WRDS download, users call `set_wrds_credentials()` to store their WRDS username and password securely in `.Renviron`; subsequent `download_data()` calls then establish the database connection automatically.

```
date_start <- as.Date("1960-01-01")
date_end <- as.Date("2024-12-31")

crsp_monthly <- download_data(
  domain = "wrds",
  dataset = "crsp_monthly",
  version = "v2",
  start_date = date_start,
  end_date = date_end,
  add_ccm_links = TRUE
)

compustat_annual <- download_data(
  domain = "wrds",
  dataset = "compustat_annual",
  start_date = date_start,
  end_date = date_end,
  additional_columns = c("at", "ib"),
```

```
  only_us = TRUE
)
```

Behind the scenes, `download_data()` performs substantial cleaning and variable construction. For CRSP monthly data (version 2), the function restricts the stock universe to common equity securities of US-incorporated corporate issuers traded on NYSE, AMEX, or NASDAQ. The function computes market capitalization as shares outstanding times price (in millions), sets zero values to missing, and creates a one-month-lagged market capitalization for portfolio weights. Exchange codes map to human-readable labels and four-digit SIC codes map to broad industry groups (e.g., Manufacturing, Finance, Utilities). The function defines excess returns as raw returns minus the Fama-French risk-free rate, floored at $-1$ to exclude implausible observations, and drops rows with missing excess returns or market capitalization. CRSP version 2 already incorporates delisting returns into the return series, eliminating the need for a separate delisting bias adjustment. Setting `add_ccm_links = TRUE` merges CRSP-Compustat linking information using link types LU and LC.

For Compustat annual data, `download_data()` restricts to industrial-format, standardized, consolidated financial statements. The function constructs book equity via a coalesce chain (stockholders' equity, then common equity plus preferred stock, then total assets minus total liabilities), adjusts for deferred taxes and investment tax credits, and subtracts preferred stock (resolved through a coalesce hierarchy of redemption, liquidating, and par values). Negative or zero book equity values become missing. The function also computes operating profitability and investment ratios from standard Compustat variables, and retains only the most recent observation per firm-year to handle duplicate filings. Setting `only_us = TRUE` keeps only USD-denominated observations.

`py-tidyfinance` provides the same `download_data()` interface in Python, so researchers can use their preferred computational environment without sacrificing functionality.

Table 1 provides a complete overview of all data sources accessible via `download_data()` in both `tidyfinance` and `py-tidyfinance`.

| Source | Description | Example |
|---|---|---|
| Ken French's Data Library | Fama-French factor models, portfolio returns, and industry classifications (169+ datasets at daily/weekly/monthly frequencies) | `download_data("famafrench", "Fama/French 5 Factors (2x3) [Daily]")` |
| Global Q | Hou-Xue-Zhang Q5 factor model (market, size, investment, ROE, expected growth) at multiple frequencies | `download_data("globalq", "factors_q5_monthly")` |

| Source | Description | Example |
|---|---|---|
| Goyal-Welch Macro Predictors | Macroeconomic predictors: dividend-price ratio, earnings-price ratio, book-to-market, term spread, default yield spread, T-bill rate, inflation | `download_data("macro_predictors", "monthly")` |
| FRED | Any economic time series from the St. Louis Fed (GDP, CPI, unemployment, etc.) | `download_data("fred", series = c("GDP", "CPIAUCNS"))` |
| Yahoo Finance | Daily OHLCV stock price data with adjusted close for any ticker | `download_data("stock_prices", symbols = c("AAPL", "MSFT"))` |
| Index Constituents | Current constituents for 21 major indexes (S&P 500, Russell 2000, DAX, MSCI World, etc.) | `download_data("constituents", index = "DAX")` |
| Open Source Asset Pricing | 213+ monthly asset pricing signals and anomaly variables | `download_data("osap")` |
| WRDS – CRSP | Monthly and daily stock returns, market cap, exchange, industry, excess returns | `download_data("wrds", "crsp_monthly")` |
| WRDS – Compustat | Annual and quarterly financial statements with customizable columns | `download_data("wrds", "compustat_annual")` |
| WRDS – CCM Links | CRSP-Compustat linking table mapping permno to gvkey | `download_data("wrds", "ccm_links")` |
| WRDS – FISD | Mergent Fixed Income Securities Database with bond characteristics | `download_data("wrds", "fisd")` |
| WRDS – TRACE | Enhanced TRACE corporate bond transactions with Dick-Nielsen cleaning | `download_data("wrds", "trace_enhanced")` |

Table 1: Data sources available via `download_data()` in `tidyfinance` and `py-tidyfinance`.

For programmatic discovery, `list_supported_types()` returns all supported datasets as a tibble, making it straightforward to explore available data without leaving the console.

```
list_supported_types() |> head(5)
```

```
# A tibble: 5 x 3
  type                  dataset_name              domain
  <chr>                 <chr>                     <chr>
```

```
1 factors_q5_daily      q5_factors_daily_2024      Global Q
2 factors_q5_weekly     q5_factors_weekly_2024     Global Q
3 factors_q5_weekly_w2w q5_factors_weekly_w2w_2024 Global Q
4 factors_q5_monthly    q5_factors_monthly_2024    Global Q
5 factors_q5_quarterly  q5_factors_quarterly_2024  Global Q
```

Similarly, `list_supported_indexes()` lists the 20 stock market indexes whose current constituents can be downloaded. Researchers can use these, for instance, to identify the member firms of a benchmark index before constructing portfolios.

```
list_supported_indexes() |> head(5)
```

```
# A tibble: 5 x 3
  index                        url                                      skip
  <chr>                        <chr>                                   <dbl>
1 DAX                          https://www.ishares.com/de/privatanleger/d~     2
2 EURO STOXX 50                https://www.ishares.com/de/privatanleger/d~     2
3 Dow Jones Industrial Average https://www.ishares.com/de/privatanleger/d~     2
4 Russell 1000                 https://www.ishares.com/ch/professionelle-
~     9
5 Russell 2000                 https://www.ishares.com/ch/professionelle-
~     9
```

## 2.2. Data cleaning and variable construction

We illustrate data cleaning and variable construction using asset growth (AG) as our running example. Asset growth is the year-over-year percentage change in total assets, $AG_t = (AT_t - AT_{t-1})/AT_{t-1}$ (Cooper, Gulen, and Schill 2008). The `add_lag_columns()` function from `tidyfinance` handles temporal alignment by creating a lagged version of total assets grouped by firm identifier (`gvkey`) with a one-year lag. Non-finite values (arising when lagged total assets are zero or missing) become `NA`. We also retain income before extraordinary items (`ib`) as a filter variable for optional exclusion of firms with negative earnings in the portfolio sort stage.

```
sorting_variable_data <- compustat_annual |>
  add_lag_columns(
    "at",
    lag = years(1),
    by = "gvkey"
  ) |>
  rename(at_lag1 = at_lag) |>
  mutate(
    sv_ag = (at - at_lag1) / at_lag1,
    sv_ag = if_else(is.finite(sv_ag), sv_ag, NA_real_),
    filter_earnings = ib
  ) |>
  select(gvkey, date, filter_earnings, sv_ag)
```

The next step merges annual sorting variables onto the monthly CRSP stock panel using

`join_lagged_values()`. The function aligns fiscal-year-end data with monthly observations through a configurable lag window. Setting `ff_adjustment = TRUE` implements the Fama-French convention (Fama and French 1993), aligning fiscal-year data from year $t$ with portfolio formation in July of year $t+1$ through a 12-to-18-month look-back window. The same lagging process generalizes to all sorting variables in the full factor library. The factor library API provides precomputed lagged panels for all variables.

Figure 2 shows how `join_lagged_values()` resolves missing observations. Given a target month $t$, the function searches backward from `min_lag` to `max_lag` and returns the most recent available value within that window.

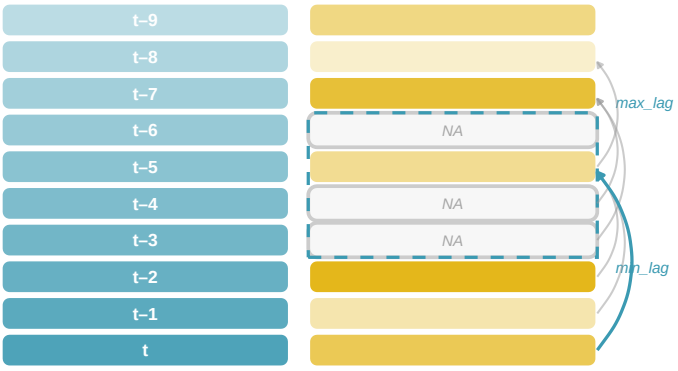



Figure 2: Illustration of the lagging mechanism in `join_lagged_values()`. Filled dots represent observed values; hollow dots represent missing observations. The function searches the window between `min_lag` and `max_lag` and selects the most recent available value (arrow).

```
sorting_data <- crsp_monthly |>
  join_lagged_values(
    sorting_variable_data,
    "gvkey",
    min_lag = months(12),
    max_lag = months(18),
    ff_adjustment = TRUE
  )
```

The resulting dataset merges monthly stock returns with the lagged sorting variable. Each row represents a stock-month observation, containing the information needed for portfolio construction: excess returns, exchange membership, industry classification, lagged market capitalization (for value-weighting), and the sorting variable itself.

```
glimpse(sorting_data)
```

```
Rows: 2,788,641
Columns: 8
$ permno          <dbl> 17326, 17326, 17326, 17326, 17326, 17326, 17326, 17326~
$ date            <date> 2022-10-01, 2022-11-01, 2022-12-01, 2023-01-01, 2023-
~
$ ret_excess      <dbl> 0.108811, -0.572900, -0.200742, 0.590241, -0.180582, -
```

```
~
$ exchange        <chr> "NASDAQ", "NASDAQ", "NASDAQ", "NASDAQ", "NASDAQ", "NAS~
$ industry        <chr> "Services", "Services", "Services", "Services", "Servi~
$ mktcap_lag      <dbl> 7.239240, 8.043600, 3.809800, 3.176991, 5.063300, 4.16~
$ filter_earnings <dbl> -31.329, -31.329, -31.329, -31.329, -31.329, -
58.625, ~
$ sv_ag           <dbl> 1.3867390, 1.3867390, 0.6142662, 0.6142662, 0.6142662,~
```

Figure 3 shows median asset growth across broad industry groups over time. Pronounced spikes during economic expansions and contractions highlight the cyclical nature of corporate investment that portfolio sorts aim to exploit.

```
sorting_data |>
  filter(!is.na(sv_ag), !is.na(industry)) |>
  mutate(year = year(date)) |>
  distinct(permno, year, .keep_all = TRUE) |>
  group_by(year, industry) |>
  summarize(median_ag = median(sv_ag, na.rm = TRUE), .groups = "drop") |>
  ggplot(aes(x = year, y = median_ag, color = industry)) +
  geom_line() +
  labs(x = NULL, y = "Median asset growth", color = "Industry") +
  theme(legend.position = "bottom")
```

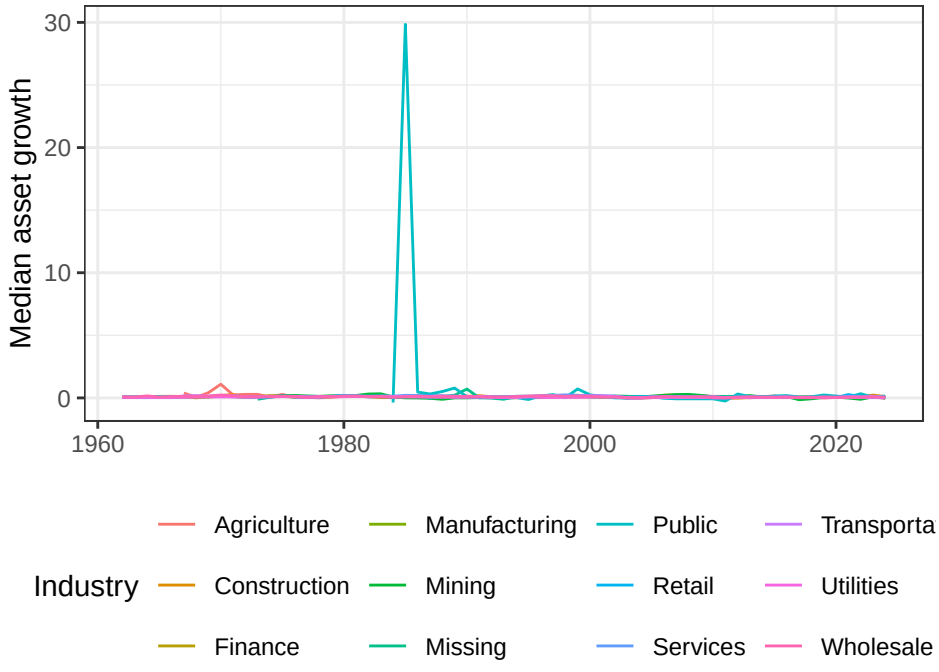



Figure 3: Median asset growth by industry group over time. Each line shows the cross-sectional median of asset growth within a broad SIC-based industry classification.

Table 2 reports cross-sectional summary statistics for asset growth across all firm-month

observations, illustrating the distribution that portfolio sorts partition into quantile portfolios.

```
sorting_data |>
  filter(!is.na(sv_ag)) |>
  summarize(
    N = n(),
    Mean = mean(sv_ag),
    SD = sd(sv_ag),
    P10 = quantile(sv_ag, 0.10),
    Median = median(sv_ag),
    P90 = quantile(sv_ag, 0.90)
  ) |>
  knitr::kable(digits = 3)
```

| N | Mean | SD | P10 | Median | P90 |
|---|---|---|---|---|---|
| 2544535 | 0.487 | 137.858 | -0.128 | 0.076 | 0.478 |

Table 2: Cross-sectional summary statistics for asset growth.

With the sorting variable constructed and merged onto the monthly stock panel, we now turn to portfolio sorts, the mechanism that translates firm-level characteristics into tradable factor returns.

## 2.3. Portfolio sorts

In portfolio sorts, stocks are ranked each period by their sorting variable and divided into quantile portfolios. Portfolio returns are equal-weighted or value-weighted averages of constituent stock returns. The long-short factor return is the difference between the extreme portfolios, for instance bottom minus top for asset growth, reflecting the negative premium documented by Cooper *et al.* (2008). Key methodological decisions include the number of portfolios, the exchange universe for computing breakpoints, the weighting scheme, and the rebalancing frequency.

The `tidyfinance` package implements this workflow through three functions. First, `breakpoint_options()` configures the sorting parameters: the number of quantile portfolios and which exchanges determine the breakpoints. Second, `compute_portfolio_returns()` takes the sorting data, applies the specified configuration, and returns portfolio-level returns. Third, `compute_long_short_returns()` constructs the hedge portfolio return from the extreme quantiles.

*Univariate sorts*

In a univariate sort, stocks are ranked solely on a single characteristic. We illustrate this with asset growth, forming decile portfolios using NYSE breakpoints and value-weighted returns, rebalanced annually in July.

```
options_main <- breakpoint_options(
```

```
  n_portfolios = 10,
  breakpoint_exchanges = "NYSE"
)
```

The `breakpoint_options()` call specifies that breakpoints come from NYSE-listed stocks only, a common convention that avoids the influence of small AMEX and NASDAQ firms on portfolio cutoffs. We then pass these options to `compute_portfolio_returns()`.

```
portfolio_returns_uni <- compute_portfolio_returns(
  sorting_data,
  "sv_ag",
  sorting_method = "univariate",
  rebalancing_month = 7,
  options_main
)
```

```
Returning a complete panel with 30 missing values in factor returns due to
insufficient observations (n() < 0).
```

```
portfolio_returns_uni |> head(10)
```

```
# A tibble: 10 x 4
   portfolio date       ret_excess_vw ret_excess_ew
       <int> <date>             <dbl>         <dbl>
 1         1 2022-10-01      0.172          0.0852
 2         1 2022-11-01      0.0102        -0.00317
 3         1 2022-12-01     -0.0705        -0.0407
 4         1 2023-01-01      0.0875         0.176
 5         1 2023-02-01     -0.0655        -0.0757
 6         1 2023-03-01     -0.0355        -0.107
 7         1 2023-04-01     -0.000521      -0.0149
 8         1 2023-05-01     -0.0701         0.00246
 9         1 2023-06-01      0.0962         0.0499
10         1 2023-07-01      0.0300         0.0382
```

The output contains monthly excess returns for each decile portfolio. To obtain the asset growth factor return, we compute the long-short portfolio using `compute_long_short_returns()`. The `direction = "bottom_minus_top"` argument means the factor is long low-AG stocks and short high-AG stocks, consistent with the finding that firms with high asset growth earn lower subsequent returns.

```
factor_returns_ag <- portfolio_returns_uni |>
  compute_long_short_returns(direction = "bottom_minus_top")

factor_returns_ag |> head(10)
```

```
# A tibble: 10 x 3
   date       ret_excess_vw ret_excess_ew
   <date>             <dbl>         <dbl>
 1 1962-04-01      NA            NA
 2 1962-05-01      NA            NA
```

```
 3 1962-06-01      NA            NA
 4 1962-07-01      -0.00202       0.00701
 5 1962-08-01      -0.0681        0.00598
 6 1962-09-01       0.0708        0.0588
 7 1962-10-01       0.0236        0.0316
 8 1962-11-01      -0.0306       -0.0232
 9 1962-12-01      -0.0154        0.0322
10 1963-01-01       0.0277        0.0106
```

*Bivariate sorts*

Bivariate (double) sorts control for a second variable, typically firm size (`mktcap_lag`), when forming portfolios. In dependent sorts, stocks are first sorted into groups by the secondary variable; within each group, they are then sorted on the main characteristic. In independent sorts, both sorts apply unconditionally, and portfolios form at the intersections. The factor return averages across the secondary dimension, isolating the effect of the primary variable after controlling for size.

```
options_main <- breakpoint_options(
  n_portfolios = 5,
  breakpoint_exchanges = "NYSE"
)
options_secondary <- breakpoint_options(
  n_portfolios = 2,
  breakpoint_exchanges = "NYSE"
)

portfolio_returns_biv <- compute_portfolio_returns(
  sorting_data,
  c("sv_ag", "mktcap_lag"),
  sorting_method = "bivariate-dependent",
  rebalancing_month = 7,
  options_main,
  options_secondary
)
```

```
Returning a complete panel with 15 missing values in factor returns due to
insufficient observations (n() < 0).
```

```
factor_returns_biv <- portfolio_returns_biv |>
  compute_long_short_returns(direction = "bottom_minus_top")

factor_returns_biv |> head(10)
```

```
# A tibble: 10 x 3
   date       ret_excess_vw ret_excess_ew
   <date>             <dbl>         <dbl>
 1 1962-04-01      NA            NA
```

```
 2 1962-05-01      NA            NA
 3 1962-06-01      NA            NA
 4 1962-07-01      -0.00972      -0.00457
 5 1962-08-01      -0.0150       -0.0133
 6 1962-09-01       0.00529       0.0258
 7 1962-10-01      -0.00836      -0.00439
 8 1962-11-01      -0.0476       -0.00611
 9 1962-12-01       0.00707       0.00238
10 1963-01-01       0.00148       0.0137
```

Dependent sorts are appropriate when the secondary variable (here, size) strongly affects the distribution of the primary sorting variable, as is typical for many accounting-based characteristics. Independent sorts treat both dimensions symmetrically and work best when both variables are of equal conceptual interest. The `sorting_method` argument supports both methods (`"bivariate-dependent"` or `"bivariate-independent"`).

*Specification grid and the factor library*

The examples above represent single points in a high-dimensional space of methodological choices. Table 3 summarizes the preprocessing decisions that can be varied when constructing factor returns. The full factor library exhaustively evaluates all valid combinations of these choices for each of the sorting variables listed in Table 4.

| Preprocessing choice | Options |
|---|---|
| Size exclusion | None; bottom 20th NYSE percentile |
| Exclusion of financials | No; Yes |
| Exclusion of utilities | No; Yes |
| Exclusion of neg. earnings | No; Yes |
| Sorting variable lag | 3 months; 6 months; FF |
| Rebalancing | Monthly; Annual |
| BP: Quantiles (main) | 5; 10 |
| Double sort | Single; Double dependent; Double independent |
| BP: Quantiles (secondary) | 2; 5 |
| BP: Exchanges | NYSE; All |
| Weighting scheme | EW; VW |

Table 3: Preprocessing choices in the portfolio sort grid. For double sorts, the secondary quantiles and exchange choices apply to the size dimension only; for univariate sorts, those rows are not applicable. FF refers to the lag procedure of Fama and French (1993).

We host precomputed factor returns for all specifications on Hugging Face and make them accessible via the `tidyfinance` API. Researchers can retrieve results without re-running the computationally intensive portfolio sort pipeline. An interactive download interface is also available via the Shiny app at https://app-download-center.cloud.sdu.dk/.

```
ag_returns <- get_available_hf_files("tidy-finance", "factor-library-grid") |>
```

```
pull(url) |>
arrow::read_parquet() |>
filter(sorting_variable == "sv_ag")
```

Section 3 provides a comprehensive overview of all available sorting variables and their empirical properties.

## 3. Factor library and API

The factor library covers 50 sorting variables spanning seven major categories: investment, profitability, valuation, momentum, intangibles, trading frictions, and financing. The selection follows the cross-section of characteristics studied by Jensen *et al.* (2023), focusing on variables with established empirical support in the asset pricing literature. Table 4 lists all sorting variables, their long-short direction, and the primary reference for each. For each variable, the full factor library provides precomputed factor returns across all specification paths defined in Table 3. We host the complete dataset on Hugging Face at https://huggingface.co/datasets/tidy-finance/factor-library. Researchers can also browse and retrieve factor returns through an interactive Shiny app at https://app-download-center.cloud.sdu.dk/.

| Variable | Name | | Main reference |
|---|---|---|---|
| 52w | 52-Week High | ↑ | George and Hwang (2004) |
| aci | Abnormal Corporate Investment | ↓ | Titman, Wei, and Xie (2004) |
| ami | Amihud Illiquidity | ↑ | Amihud (2002) |
| ag | Asset Growth | ↓ | Cooper *et al.* (2008) |
| beta | Beta | ↑ | Fama and MacBeth (1973) |
| bm | Book-to-Market | ↑ | Daniel and Titman (2006) |
| cfm | Cash Flow to Market | ↑ | Lakonishok, Shleifer, and Vishny (1994) |
| cbop | Cash-Based Operating Profitability | ↑ | Ball, Gerakos, Linnainmaa, and Nikolaev (2016) |
| dbe | Change in Common Equity | ↓ | Richardson, Sloan, Soliman, and Tuna (2005) |
| dcol | Change in Current Operating Liabilities | ↓ | Richardson *et al.* (2005) |
| dfnl | Change in Financial Liabilities | ↓ | Richardson *et al.* (2005) |
| dnoa | Change in Net Operating Assets | ↓ | Hirshleifer, Hou, Teoh, and Zhang (2004) |
| dpia | Change in PPE and Intangibles | ↓ | Lyandres, Sun, and Zhang (2008) |
| dwc | Change in Working Capital | ↓ | Richardson *et al.* (2005) |
| cdi | Composite Debt Issuance | ↓ | Lyandres *et al.* (2008) |
| csi | Composite Share Issuance | ↓ | Daniel and Titman (2006) |
| dm | Debt to Market | ↑ | Bhandari (1988) |
| dtv | Dollar Trading Volume | ↓ | Brennan, Chordia, and Subrahmanyam (1998) |
| em | Earnings to Market | ↑ | Basu (1983) |
| ebm | Enterprise Book to Market | ↑ | Penman, Richardson, and Tuna (2007) |

| | | | |
|---|---|---|---|
| gpa | Gross Profit Asset | ↑ | Novy-Marx (2013) |
| hr | Hiring Rate | ↓ | Belo and Lin (2012) |
| iskew | Idiosyncratic Skewness | ↓ | Bali, Engle, and Murray (2016) |
| ivol | Idiosyncratic Volatility | ↓ | Ang, Hodrick, Xing, and Zhang (2006) |
| dinv | Inventory Changes | ↓ | Thomas and Zhang (2002) |
| ig | Investment Growth | ↓ | Xing (2008) |
| rev | Long-Term Reversal | ↓ | De Bondt and Thaler (1985) |
| me | Market Equity | ↓ | Banz (1981) |
| mdr | Maximum Daily Return | ↓ | Bali, Cakici, and Whitelaw (2011) |
| mom | Momentum | ↑ | Fama and French (1996) |
| ndf | Net Debt Financing | ↓ | Baker and Wurgler (2006) |
| ndm | Net Debt to Market | ↓ | Penman *et al.* (2007) |
| nef | Net Equity Financing | ↓ | Baker and Wurgler (2006) |
| nxf | Net External Financing | ↓ | Baker and Wurgler (2006) |
| noa | Net Operating Assets | ↓ | Hirshleifer *et al.* (2004) |
| npy | Net Payout Yield | ↑ | Boudoukh, Michaely, Richardson, and Roberts (2007) |
| o | O-Score | ↓ | Ohlson (1980); Dichev (1998) |
| oa | Operating Accruals | ↓ | Sloan (1996) |
| ocm | Operating Cash Flow to Market | ↑ | Desai, Rajgopal, and Venkatachalam (2004) |
| ope | Operating Efficiency | ↑ | Fama and French (2015) |
| ol | Operating Leverage | ↑ | Novy-Marx (2011) |
| rer | Real-Estate Ratio | ↑ | Titman *et al.* (2004) |
| rs | Relative Strength | ↑ | Jegadeesh and Livnat (2006) |
| rmom | Residual Momentum | ↑ | Blitz, Huij, and Martens (2011) |
| roa | Return on Assets | ↑ | Balakrishnan, Bartov, and Faurel (2010) |
| roe | Return on Equity | ↑ | Hou, Xue, and Zhang (2014) |
| sm | Sales to Market | ↑ | Barbee Jr, Mukherji, and Raines (1996) |
| tur | Share Turnover | ↓ | Datar, Naik, and Radcliffe (1998) |
| sue | Standardized Unexpected Earnings | ↑ | Foster, Olsen, and Shevlin (1984) |
| tbi | Taxable Income to Book Income | ↓ | Lev and Nissim (2004) |

Table 4: Sorting variables in the factor library. The direction column indicates whether the long-short factor return goes long top-minus-bottom (↑) or bottom-minus-top (↓) of the characteristic distribution.

```
Warning: There was 1 warning in `mutate()`.
i In argument: `sorting_variable = forcats::fct_reorder(...)`.
Caused by warning:
! `fct_reorder()` removing 9980 missing values.
i Use `.na_rm = TRUE` to silence this message.
i Use `.na_rm = FALSE` to preserve NAs.
```

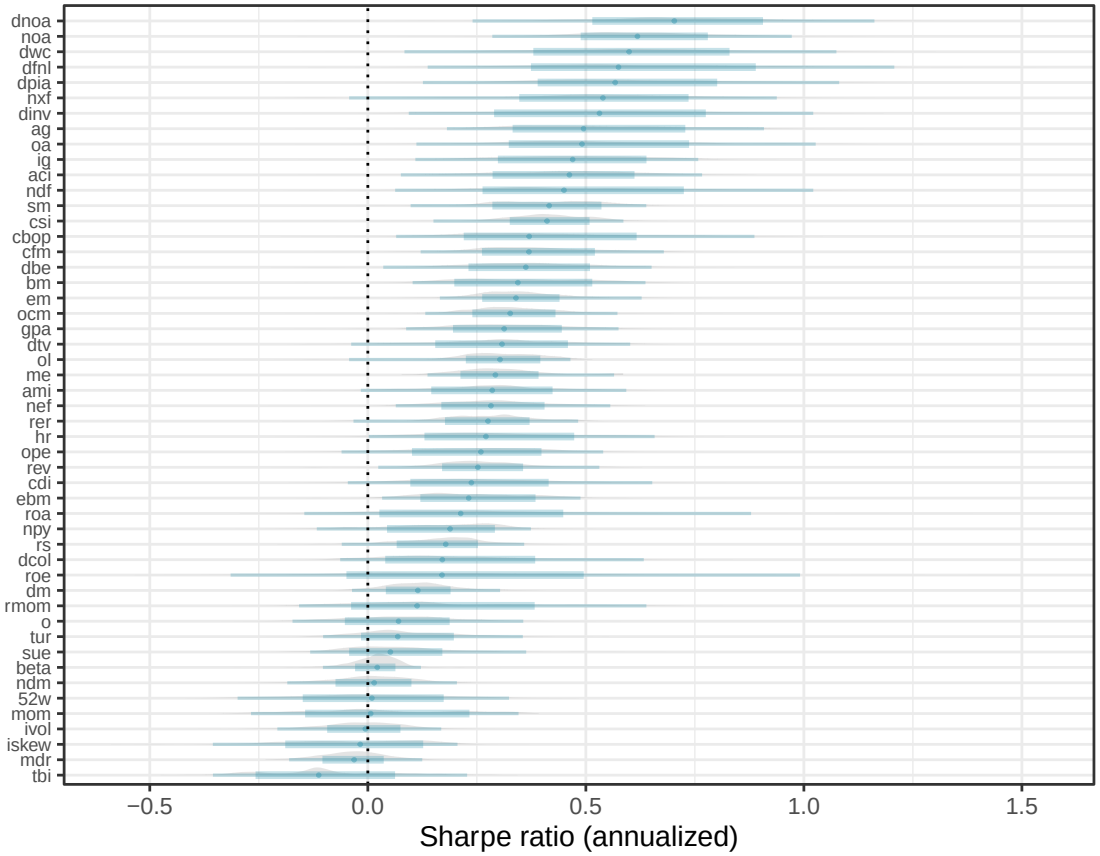



Figure 4: **Sharpe ratios.** The ridge plots indicate the distribution of annualized Sharpe ratios of long-short return differentials for a given sorting variable across all data preprocessing choices illustrated in Table 3. The estimates are based on the available sample period for each factor.

```
Warning: Removed 9980 rows containing missing values or values outside the scale range
(`stat_slabinterval()`).
```

# 4. Summary and discussion

# Computational details

We obtained the results in this paper using R~4.5.2 with the following packages: **tidyfinance**~0.4.5.9017, **dplyr**~1.1.4, **tidyr**~1.3.1, **ggplot2**~4.0.0, **lubridate**~1.9.4, **arrow**~21.0.0.1, **stringr**~1.5.2, **forcats**~1.0.0, **ggdist**~3.3.3, and **knitr**~1.51. CRAN hosts R and all packages at https://CRAN.R-project.org/. `renv`~1.1.5 ensures full reproducibility of the computational environment.

We publish all code, data workflows, and documentation at https://github.com/tidy-finance/jss-multilingual-factor-library.

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# Technical details

Internet Appendix

Variable construction

In this section, we show the construction details for the risk factors. We group them by their underlying mechanism as shown in Table 4 in the Appendix of the main paper.

# IA-1. Financing

### CDI - Composite Debt Issuance

Following Lyandres *et al.* (2008), we compute composite debt issuance as the logarithmic growth rate of book debt (i.e., current debt, DLC, plus long-term debt, DLTT) over the last five years, i.e.,

$$\text{CDI}_t = \log(\text{DLC}_t + \text{DLTT}_t) - \log(\text{DLC}_{t-60} + \text{DLTT}_{t-60})$$

We replace the missing values of DLC and DLTT with zero individually. Additionally, the sum of DLC and DLTT must be larger than zero, and is set to missing otherwise.

### CSI - Composite Share Issuance

Following Daniel and Titman (2006), we measure composite share issuance from CRSP data as the difference between the change in market equity (ME) (i.e., the shares outstanding, shrout, times the absolute value of their price, altprc, in million USD) and the cumulative log return (where $r$ is a simple return) in each month from $t$ to $t-60$, i.e.,

$$\text{CSI}_t = \log(ME_t/ME_{t-60}) - \sum_{\tau=t-59}^{t} \log(1+r_\tau)$$

### DBE - Change in Common Equity

Following Richardson *et al.* (2005), we calculate the change in common equity as the change in common equity (CEQ) over one year scaled by lagged assets (AT), i.e.,

$$DBE_t = \frac{CEQ_t - CEQ_{t-12}}{AT_{t-12}}$$

### DCOL - Change in Current Operating Liabilities

Following Richardson *et al.* (2005), we measure the change in current operating liabilities as the change in current operating liabilities (i.e., current liabilities, LCT, minus current debt, DLC) over one year scaled by lagged assets (AT), i.e.,

$$DCOL_t = \frac{(LCT_t - DLC_t) - (LCT_{t-12} - DLC_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC with 0.

**DFNL - Change in Financial Liabilities**

Following Richardson *et al.* (2005), we measure the change in financial liabilities (i.e., the sum of current debt, DLC, plus long-term debt, DLTT, plus preferred stocks, PSTK) over one year scaled by lagged assets (AT), i.e.,

$$DFNL_t = \frac{(DLC_t + DLTT_t + PSTK_t) - (DLC_{t-12} + DLTT_{t-12} + PSTK_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC, DLTT and PSTK with 0, if at least one of them is given.

**NDF - Net Debt Financing**

Following Baker and Wurgler (2006), we measure net debt financing as cash proceeds from the issuance of long-term debt (DLTIS) minus cash payments for long-term debt reductions (DLTR) plus net changes in current debt (DLCCH) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NDF_t = \frac{DLTIS_t - DLTR_t + DLCCH_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

We replace the missing values of DLCCH with zero.

**NEF - Net Equity Financing**

Following Baker and Wurgler (2006), we measure net equity financing as proceeds from the sale of common and preferred stocks (SSTK) minus payments for the repurchase of common and preferred stocks (PRSTKC) minus cash payments for dividends (DV) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NEF_t = \frac{SSTK_t - PRSTKC_t - DV_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

**NXF - Net External Financing**

Following Baker and Wurgler (2006), we measure net external financing as the sum of net debt financing (NDF) and net equity financing (NEF) as defined above.

## IA-2. Intangibles

**HR - Hiring Rate**

Following Belo and Lin (2012), we measure the hiring rate as the change in the number of employees (EMP) over one year scaled by the average number of employees, i.e.,

$$HR_t = \frac{EMP_t - EMP_{t-12}}{\frac{1}{2}(EMP_t + EMP_{t-12})}$$

We exclude firms with a hiring rate of zero.

**OL - Operating Leverage**

Following Novy-Marx (2011), we measure operating leverage as cost of goods sold (COGS) plus selling, general, and administrative expenses (XSGA), both scaled by current total assets (AT), i.e.,

$$OL_t = \frac{COGS_t + XSGA_t}{AT_t}$$

**RER - Real-Estate Ratio**

Following Titman *et al.* (2004), we measure the real-estate ratio in two subperiods differently.

(a) Prior to 1983, the real-estate ratio is the sum of buildings (PPENB) and capital leases (PPENLS) scaled by net property, plant, and equipment (PPENT), i.e.,

$$RER_t = \frac{PPENB_t + PPENLS_t}{PPENT_t}$$

(b) After the end of 1983, the real-estate ratio is the sum of buildings at cost (FATB) and leases at cost (FATL), both divided by gross property, plant, and equipment (PPEGT), i.e.,

$$RER_t = \frac{FATB_t + FATL_t}{PPEGT_t}$$

Subsequently, we winsorize the real-estate ratios in each fiscal year at the first and 99th percentile. The industry-adjusted real-estate ratio is obtained by subtracting the industry average real-estate ratio from each stock-specific real-estate ratio. We use 2-digit SIC codes to assign stocks to industries. We always require at least five observations to calculate the industry average each year.

## IA-3. Investment

**ACI - Abnormal Corporate Investment**

Following Titman *et al.* (2004), we measure abnormal corporate investments using the variable CE, which is capital expenditures (CAPX) scaled by sales (SALE), then ACI is given by CE scaled by CE from the previous three years minus one, i.e.,

$$ACI_t = \frac{CE_t}{\frac{1}{3}(CE_{t-12} + CE_{t-24} + CE_{t-36})} - 1$$

We follow Hou, Xue, and Zhang (2020) and exclude stocks with sales below 10 million dollars.

**AG - Asset Growth**

Following Cooper *et al.* (2008), we measure asset growth as the change in total assets (AT) over one year scaled by lagged assets (AT), i.e.,

$$AG_t = \frac{AT_t - AT_{t-12}}{AT_{t-12}}$$

**DINV - Inventory Changes**

Following Thomas and Zhang (2002), we measure the change in inventory as the annual change in inventories (INVT) scaled by average total assets (AT) over the current and the last year, i.e.,

$$DINV_t = \frac{INVT_t - INVT_{t-12}}{\frac{1}{2}(AT_t + AT_{t-12})}$$

**DNOA - Change in Net Operating Assets**

Following Hirshleifer *et al.* (2004), we measure net operating assets (defined as $NOA^{int}$) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, the change in net operating assets is the difference between net operating assets of fiscal year $t$ and fiscal year $t-12$ scaled by lagged assets (AT), i.e.,

$$DNOA_t = \frac{NOA_t^{int} - NOA_{t-12}^{int}}{AT_{t-12}}$$

**DPIA - Change in Property, Plant, Equipment, and Inventory**

Following Lyandres *et al.* (2008), we measure the change in property, plant, equipment, and inventory by combining the change in gross property, plant, and equipment (PPEGT) with the annual change in inventory (INVT) and scaling this sum by lagged total assets (AT), i.e.,

$$DPIA_t = \frac{(PPEGT_t - PPEGT_{t-12}) + (INVT_t - INVT_{t-12})}{AT_{t-12}}$$

**DWC - Change in Net Non-Cash Working Capital**

Following Richardson *et al.* (2005), we measure non-cash working capital as

$$WC_t = (ACT_t - CHE_t) - (LCT_t - DLC_t)$$

where ACT corresponds to current assets, CHE to cash, LCT to current liabilities, and DLC to short-term debt. We set missing values of DLC to zero. Then, the change in net non-cash

working capital corresponds to the change in $WC$ from fiscal year $t-12$ to fiscal year $t$ scaled by lagged total assets (AT), i.e.,

$$DWC_t = \frac{WC_t - WC_{t-12}}{AT_{t-12}}$$

**IG - Investment Growth**

Following Xing (2008), we measure investment growth as the annual change in capital expenditures (CAPX) scaled by lagged capital expenditures, i.e.,

$$IG_t = \frac{CAPX_t - CAPX_{t-12}}{CAPX_{t-12}}$$

**NOA - Net Operating Assets**

Following Hirshleifer *et al.* (2004), we measure net operating assets (defined as $NOA^{int}$) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, net operating assets is the net operating assets of fiscal year $t$ scaled by lagged assets (AT), i.e.,

$$NOA_t = \frac{NOA_t^{int}}{AT_{t-12}}$$

**OA - Operating Accruals**

Following Sloan (1996) (before 1988) and Hribar and Collins (2002) (after 1988), we measure operating accruals until 1988 as

$$OA_t = \frac{(\Delta ACT_t - \Delta CHE_t) - (\Delta LCT_t - \Delta DLC_t - \Delta TXP_t) - DP_t}{AT_{t-12}}$$

where ACT is current assets, CHE cash, LCT current liabilities, DLC short-term debt, TXP taxes payable, and DP depreciation and amortization. Moreover, we replace the missing values of DLC and TXP with zero. $\Delta$ represents changes over one year. From 1988 onwards, we measure operating accruals as

$$OA_t = \frac{NI_t - OANCF_t}{AT_{t-12}}$$

where NI is net income and OANCF corresponds to net cash flow from operations.

# IA-4. Momentum

### MOM - Return Momentum

Following Fama and French (1996), we measure return momentum from monthly CRSP returns (where $r$ is a simple return) as the cumulative return from month $t-11$ to month $t-1$ skipping the most recent month, i.e.,

$$\text{MOM}_t = \exp\left(\sum_{\tau=t-11}^{t-1} \log(1+r_\tau)\right) - 1$$

### RMOM - Residual Momentum

Following Blitz *et al.* (2011), we define the 11-month residual momentum (RMOM) in each month and for each stock as cumulative residual returns from month $t-11$ to month $t-1$, scaled by the standard deviation of residual returns over the same time horizon. Residual returns $\epsilon$ are obtained in each month from regressing monthly simple, excess stock returns from month $t-1$ to month $t-36$ on the Fama and French (1993) three-factor model. Throughout these rolling regressions, we always require 36 monthly returns.

### RS - Revenue Surprise

Following Jegadeesh and Livnat (2006), we measure the change in revenues per share (DRPS) from quarterly Compustat data as quarterly sales (SALEQ) scaled by the product of quarterly shares outstanding (CSHPRQ) times their cumulative adjustment factor (AJEXQ), i.e.,

$$DRPS_q = \frac{SALEQ_q}{CSHPRQ_q \cdot AJEXQ_q} - \frac{SALEQ_{q-4}}{CSHPRQ_{q-4} \cdot AJEXQ_{q-4}}$$

Then, the revenue surprise corresponds to the change in revenues per share over the last four quarters scaled by the standard deviation of the change in revenues per share over the last eight quarters. We require at least six quarterly observations for this rolling standard deviation and remove infinite values. Finally, the earnings announcement date has to be after the fiscal quarter end date.

### SUE - Standardized Unexpected Earnings

Following Foster *et al.* (1984), we measure unexpected earnings per share (UES) from quarterly Compustat data as the change in split-adjusted earnings (which are equal to quarterly earnings per share, EPSPXQ, scaled by the correction factor for shares outstanding, AJEXQ) per share from its value four quarters ago

$$UES_q = \frac{EPSPXQ_q}{AJEXQ_q} - \frac{EPSPXQ_{q-4}}{AJEXQ_{q-4}}$$

Then, standardized unexpected earnings are defined as unexpected earnings per share divided by the standard deviation of unexpected earnings per share over the previous eight quarters. We require at least six quarterly observations for this rolling standard deviation. Finally, the earnings announcement date has to be after the fiscal quarter end date.

**52W - 52-Week High**

Following George and Hwang (2004), we measure the 52-week high as month $t$'s closing split-adjusted stock price scaled by the highest daily split-adjusted stock price over the previous twelve months. We only consider stock-month pairs with more than three valid price observations.

**REV - Long-Term Reversal**

Following De Bondt and Thaler (1985), we measure the long-term reversal effect by the cumulative returns from month $t-60$ to month $t-13$, i.e.,

$$\text{REV}_t = \exp\left(\sum_{\tau=t-60}^{t-13} \log(1+r_\tau)\right) - 1$$

## IA-5. Profitability

**CBOP - Cash-Based Operating Profitability**

Following Ball *et al.* (2016), we measure cash-based operating profitability

$$\begin{aligned} CBOP_t =& (REVT_t - COGS_t - XSGA_t + XRD_t \\ & - \Delta RECT_t - \Delta INVT_t - \Delta XPP_t + \Delta DRC_t \\ & + \Delta DRLT_t + \Delta AP_t + \Delta XACC_t)/AT_t \end{aligned}$$

where REVT is total revenue, COGS are cost of goods sold, XSGA are selling, general, and administrative expenses, and XRD are R&D expenses. Moreover, $\Delta$ refers to one-year changes, which we measure for accounts receivable (RECT), inventory (INVT), prepaid expenses (XPP), deferred revenues (the sum of current, DRC, and long-term, DRLT), trade accounts payable (AP), and accrued expenses (XACC). We set the missing values of XRD and all missing changes to zero.

**GPA - Gross Profits to Assets**

Following Novy-Marx (2013), we measure gross profits to assets as total revenues (REVT) minus cost of goods sold (COGS) scaled by current total assets (AT), i.e.,

$$GPA_t = \frac{REVT_t - COGS_t}{AT_t}$$

**O - Ohlson's O-score**

Following Ohlson (1980), we measure O-score with the following linear relation

$$
\begin{aligned}
O_t = &- 1.32 - 0.407 \cdot \log(AT_t) + 6.03 \cdot \frac{DLC_t + DLTT_t}{AT_t} \\
&- 1.43 \cdot \frac{ACT_t - LCT_t}{AT_t} + 0.076 \cdot \frac{LCT_t}{AT_t} \\
&- 1.72 \cdot \mathbf{1}_{LT_t > AT_t} - 2.37 \cdot \frac{NI_t}{AT_t} - 1.83 \cdot \frac{PI_t + DP_t}{LT_t} \\
&+ 0.285 \cdot \mathbf{1}_{NI_t<0 \text{ and } NI_{t-12}<0} - 0.521 \cdot \frac{NI_t - NI_{t-12}}{|NI_t| + |NI_{t-12}|}
\end{aligned}
$$

where AT corresponds to total assets, DLC to short-term debt, DLTT to long-term debt, ACT to current assets, LCT to current liabilities, LT to total liabilities, PI to pretax income, DP to depreciation and amortization, and NI to net income. We winsorize all variables except for dummy variables at the first and 99$^{\text{th}}$ percentile of their respective distribution.

**OPE - Operating Profits to Book Equity**

Following Fama and French (2015), we measure operating profits to book equity as

$$
OPE_t = \frac{REVT_t - COGS_t - XSGA_t - XINT_t}{BE_t}
$$

where REVT corresponds to total revenues, COGS to cost of goods sold, XSGA to selling, general and administrative expenses, XINT to interest expenses, and BE to book equity. The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero. Moreover, missing values in COGS, XSGA, and XINT are set to zero.

**ROA - Return on Assets**

Following Balakrishnan *et al.* (2010), we measure return on assets as quarterly income before extraordinary items (IBQ) relative to lagged quarterly total assets (ATQ), i.e.,

$$
ROA_q = \frac{IBQ_q}{ATQ_{q-1}}
$$

Finally, the earnings announcement date has to be after the fiscal quarter end date.

**ROE - Return on Equity**

Following Hou *et al.* (2014), we measure return on equity as quarterly income before extraordinary items (IBQ) relative to lagged book equity (BEQ)

$$ROE_q = \frac{IBQ_q}{BEQ_{q-1}}$$

Quarterly book equity (BEQ) is computed as the (a) book equity of shareholders plus (b) balance sheet deferred taxes and investment tax credit minus (c) the book value of preferred stock. Regarding (a), we measure book equity of shareholders by shareholders' equity (SEQQ), or the sum of common equity (CEQQ) and the par value of preferred stock (PSTKQ), or if all previous items are unavailable by total assets (ATQ) minus total liabilities (LTQ). Regarding (b), the balance sheet deferred taxes and investment tax credit is TXDITCQ, or TXDBQ, or zero if both are missing. Regarding (c), the book value of preferred stocks corresponds to PSTKRQ, or PSTKQ, or zero if both are unavailable. Finally, the earnings announcement date has to be after the fiscal quarter end date.

**TBI - Taxable Income to Book Income**

Following Green, Hand, and Zhang (2013), we measure taxable income to book income as pretax income (PI) scaled by net income (NI)

$$TBI_t = \frac{PI_t}{NI_t}$$

We require positive pretax and net income.

## IA-6. Size

**ME - Size**

Following Fama and French (1992), we measure size as the natural logarithm of the market equity. We obtain market equity data from CRSP by multiplying the shares outstanding (shrout * 1,000, as they are given in thousands) with the corresponding share price (the absolute value of altprc), i.e.,

$$ME_t = \log\left(shrout_t * 1000 * |altprc_t|\right)$$

## IA-7. Trading frictions

**AMI - Amihud Illiquidity Measure**

Following Amihud (2002), we measure the illiquidity from daily return-to-volume (RV), which is the absolute daily return (r) scaled by the daily dollar trading volume (product of vol and prc), i.e.,

$$RV_t = \frac{|r_d|}{prc_t * vol_t}$$

Then, the Amihud illiquidity measure corresponds to the average return to volume estimate over the last six months. We require at least 50 observations for this average and adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

**BETA - Beta Relative to the Market**

Following Fama and MacBeth (1973), we measure the market beta $\beta_1$ from monthly CRSP data. Specifically, we run the following time series regression over the previous five years

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + u_t$$

Moreover, we require at least 24 monthly observations for the regression above.

**DTV - Dollar Trading Volume**

Following Brennan *et al.* (1998), we measure dollar trading volume from daily CRSP data as the average dollar trading volume (product of vol and prc) from month $t-6$ to month $t-1$ (with at least 50 observations). Moreover, we adjust dollar trading volume from NASDAQ according to Gao and Ritter (2010).

**ISKEW - Idiosyncratic Skewness Relative to the Fama and French (1993) three-factor model**

Following Bali *et al.* (2016), we regress daily excess returns in each month on the three Fama and French (1993) factor model (with a minimum of 15 daily observations), i.e.,

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

Then, idiosyncratic skewness is measured as the skewness of residuals $u_t$.

**IVOL - Idiosyncratic Volatility Relative to the Fama and French (1993) model**

Following Ang *et al.* (2006), we compute idiosyncratic volatility relative to the Fama and French (1993) three-factor model as the volatility of residuals from the following regression (with a minimum of 15 daily observations)

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

**MDR - Maximum Daily Return**

Following Bali *et al.* (2011), we measure the maximum daily (excess) return in each month from daily CRSP data (with at least 15 return observations).

**TUR - Share Turnover**

Following Datar *et al.* (1998), we measure the daily share turnover of each stock as the number of shares traded (vol) scaled by the number of shares outstanding (shrout), i.e.,

$$T_t = \frac{vol}{shrout}$$

Then, the share turnover for each firm in each month $t$ is the average daily share turnover over the previous six months (with at least 50 observations). Moreover, we adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

## IA-8. Valuation

**BM - Book Equity to Market Equity**

Following Davis, Fama, and French (2000), we measure the book-to-market ratio as book equity (BE) from Compustat divided by market equity (ME) from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$BM_t = \frac{BE_t}{ME_t}$$

The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero.

**CFM - Cash Flow to Market Equity**

Following Lakonishok *et al.* (1994), we measure the cash-flow-to-price ratio as income before extraordinary items (IB) plus depreciation (DP) scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$CFM_t = \frac{IB_t + DP_t}{ME_t}$$

We exclude all stocks with negative cash flows.

**DM - Debt to Market Equity**

Following Bhandari (1988), we measure the debt-to-market ratio as short-term debt (DLC) plus long-term debt (DLTT) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$DM_t = \frac{DLC_t + DLTT_t}{ME_t}$$

We exclude stocks with missing DLC and DLTT observations.

**EBM - Enterprise Book Equity to Market Equity**

Following Penman *et al.* (2007), we measure enterprise book equity scaled by market equity as net debt plus book equity scaled by net debt plus market equity

$$EBM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + BE_t}{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments, and ME to the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD). Lastly, in this case, book equity BE is computed as common equity (CEQ) plus TSTKP minus DVPA. Missing observations in DVPA and TSTKP are set to zero. Additionally, we require that the sum of net debt and book equity, as well as the sum of net debt plus market equity, are positive.

**EM - Earnings to Market Equity**

Following Basu (1983), we measure the earnings-to-price ratio as income before extraordinary items (IB) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$EM_t = \frac{IB_t}{ME_t}$$

We exclude firms with negative earnings.

**NDM - Net Debt to Market Equity**

Following Penman *et al.* (2007), we measure net debt to price as net debt from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$NDM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t}{ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments. Lastly, missing observations in DVPA and TSTKP are set to zero.

**NPY - Net Payout Yield**

Following Boudoukh *et al.* (2007), we measure the net payout yield as net payouts from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$\begin{aligned} NPY_t = &[(DVC_t + PRSTKC_t + \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV<0}) \\ &- (SSTK_t - \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV>0})]/ME_t \end{aligned}$$

where DVC are dividends from common stock, PRSTKC is the purchase of common and preferred stock, PSTKRV is the value of the net number of preferred stocks outstanding, and SSTK reflects the sale of common and preferred stocks. $1_{\Delta PSTKRV<0}$ is a dummy variable that has value one if the annual change in PSTKRV is negative and zero otherwise. Moreover, we exclude stocks with negative net payouts.

**OCM - Operating Cash Flow to Market Equity**

Following Desai *et al.* (2004), we measure the ratio of operating cash flows to price as operating cash flows (OC) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$OCM_t = \frac{OC_t}{ME_t}$$

Before 1988, we measure OC as funds from operations (FOPT) minus the change in working capital (item WCAP). Thereafter, we measure OC as net cash flows from operating activities (OANCF). Moreover, we exclude firms with negative operating cash flows.

**SM - Sales to Market Equity**

Following Barbee Jr *et al.* (1996), we measure the sales-to-price ratio (SM) as sales (SALE) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$SM_t = \frac{SALE_t}{ME_t}$$

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